PROJECTIVE DIMENSION IS ADDITIVE ALONG A QUOTIENT RING OF FINITE PROJECTIVE DIMENSION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot; nothing was added, removed or redefined and no package was added.

Editorial note. The source URL is the source-level Stacks Project URL, as required when more than one card is produced from one source; the per-claim tag is carried by claim_key.

Editorial note. The author writes 'the same (this is trivial but also follows from Lemma ...)' in the counted proof span; that aside is reproduced verbatim, including its missing full stop. The four references inside the counted proof span are recorded as non-load-bearing navigation pointers.

Allowed standard prerequisites (editorial).

- Finite projective dimension for a finite module over a Noetherian ring is equivalent to being perfect. invoked at proof lines 29872-29874. Connects the hypothesis to the perfectness language and provides finiteness over R.
- The Auslander-Buchsbaum formula pd + depth = depth of the ring. invoked at proof lines 29877-29886. Supplies the three displayed equations whose combination gives additivity.
- Equality of the depth of a finite module over the ring and over a quotient. used at proof lines 29887-29890. Allows the three equations to be combined consistently.

Source reference. Original lemma 0AUK: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 13653–13653. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 090V: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11423–11423. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original definition 18871: sources/stacks-more-algebra/pinned-original-source.tex, source lines 18871–18871. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 18987: sources/stacks-more-algebra/pinned-original-source.tex, source lines 18987–18987. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Lemma 0.2. *Let R be a Noetherian local ring. Let $I \subset R$ be an ideal and let E be a nonzero finite module over R/I. If R/I has finite projective dimension and E has finite projective dimension over R/I, then E has finite projective dimension over R and*

$$pd_R(E) = pd_R(R/I) + pd_{R/I}(E)$$

Proof. We will use that, for a finite module, having finite projective dimension over R, resp. R/I is the same as being a perfect module, see discussion following Definition 18871. We see that E has finite projective dimension over R by Lemma 18987. Thus we can apply Auslander-Buchsbaum (Algebra, Proposition 090V) to see that

$$\mathrm{pd}_R(E) + \mathrm{depth}(E) = \mathrm{depth}(R), \quad \mathrm{pd}_{R/I}(E) + \mathrm{depth}(E) = \mathrm{depth}(R/I),$$

and

$$\mathrm{pd}_R(R/I) + \mathrm{depth}(R/I) = \mathrm{depth}(R)$$

Note that in the first equation we take the depth of E as an R-module and in the second as an R/I-module. However these depths are the same (this is trivial but also follows from Algebra, Lemma 0AUK). This concludes the proof. □